CORE PURE / FURTHER CALCULUS

Integration by Parts

Questions by topic 15 questions · 105 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Instructions

Integration by parts is assumed knowledge from A Level Mathematics. This pack revises it at Further Mathematics level with new functions that A-level Further Mathematics introduces.

Questions

Question 1	*5 marks*	2
Question 2	*6 marks*	3
Question 3	*7 marks*	4
Question 4	*5 marks*	5
Question 5	*7 marks*	6
Question 6	*7 marks*	7
Question 7	*7 marks*	8
Question 8	*9 marks*	9
Question 9	*9 marks*	10
Question 10	*6 marks*	11
Question 11	*6 marks*	12
Question 12	*6 marks*	13
Question 13	*6 marks*	14
Question 14	*7 marks*	15
Question 15	*12 marks*	16

1. Use integration by parts to find

$$\int_1^9 \sqrt{x} \ln\left(\frac{9}{x}\right) \mathrm{d}x$$

giving your answer in the form $a + b\ln 3$, where a and b are rational constants to be found. **[5]**

2. Given that

$$y = \arctan\left(\frac{x-2}{x+2}\right)$$

where $x \neq -2$

(a) find $\frac{\mathrm{d}y}{\mathrm{d}x}$, giving your answer in its simplest form. **[2]**

(b) Use integration by parts to find

$$\int \arctan\left(\frac{x-2}{x+2}\right) \mathrm{d}x$$ **[4]**

3. Use integration by parts to find

$$\int_0^2 x^2 \ln(1+x)\,\mathrm{d}x$$

giving your answer in the form $a + b\ln 3$, where a and b are rational constants to be found. **[7]**

4. Use integration by parts to find

$$\int \operatorname{arsinh}(2x+1)\,\mathrm{d}x$$

giving your answer in terms of logarithms. **[5]**

5.

$$h(\theta) = (3\sin\theta - \cos\theta)(\sin\theta + 2\cos\theta)$$

(a) Show that $h(\theta) = \frac{1}{2} - \frac{5}{2}\cos 2\theta + \frac{5}{2}\sin 2\theta$. **[2]**

(b) Hence, using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta\right) h(\theta)\,\mathrm{d}\theta$$ **[5]**

6. **(a)** Find

$$\int x^2 \arctan x \, \mathrm{d}x \qquad \textbf{[5]}$$

(b) Hence find the exact value of

$$\int_0^1 x^2 \arctan x \, \mathrm{d}x \qquad \textbf{[2]}$$

7. **(a)** Find

$$\int x^3 \arctan x \, \mathrm{d}x$$ **[5]**

(b) Hence find the exact value of

$$\int_0^{\sqrt{3}} x^3 \arctan x \, \mathrm{d}x$$ **[2]**

8. **(a)** Using the logarithmic form of $\operatorname{artanh} x$, prove that the derivative of $\operatorname{artanh} x$ is

$$\frac{1}{1-x^2}$$

for $|x| < 1$. **[3]**

(b) Hence find

$$\int_0^{1/2} \operatorname{artanh} x \, \mathrm{d}x$$

giving your answer in exact logarithmic form. **[5]**

(c) Priya tries to evaluate

$$\int_0^{3/2} \operatorname{artanh} x \, \mathrm{d}x$$

using her calculator, and gets an 'error'. Explain why. **[1]**

9. **(a)** Using the logarithmic form

$$\operatorname{artanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \qquad |x| < 1$$

prove that the derivative of $\operatorname{artanh} x$ is

$$\frac{1}{1-x^2}$$

[3]

(b) Hence find

$$\int_0^{1/3} \operatorname{artanh}(2x) \, \mathrm{d}x$$

giving your answer in exact logarithmic form. **[5]**

(c) Noah tries to evaluate

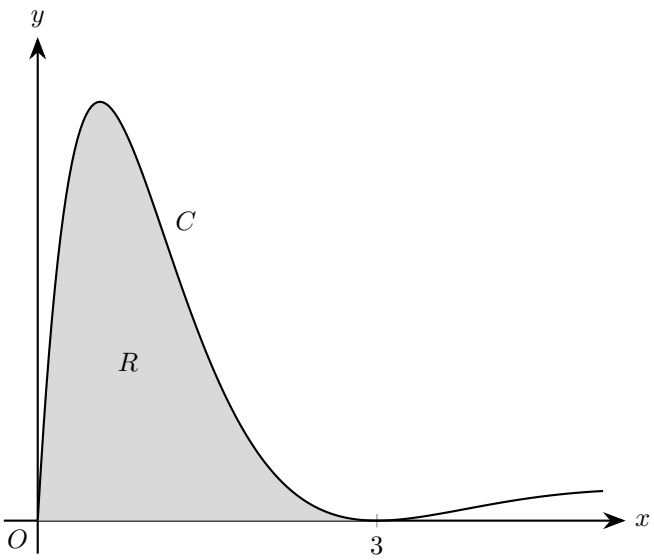
$$\int_0^{1/2} \operatorname{artanh}(2x) \, \mathrm{d}x$$

using his calculator, and gets an error. Explain why. **[1]**

10. For $0 < x < 1$, use integration by parts to find

$$\int x \operatorname{artanh}(2x-1)\,\mathrm{d}x$$

giving your final answer in terms of logarithms. **[6]**



11. The sketch shows part of the curve C given by

$$y = x(3-x)^2e^{-x} \qquad x \geq 0$$

The shaded region R is bounded by C and the x-axis for $0 \leq x \leq 3$.

Find the exact area of R, giving your answer in the form

$$A + Be^{-3}$$

where A and B are rational numbers. **[6]**

12. Show that

$$\int_{e^{-1/2}}^{1} x(\ln x)^2 \, \mathrm{d}x = a + \frac{b}{e}$$

where a and b are rational constants to be found. **[6]**

13. Given that

$$y = \arcsin\left(\frac{x-1}{3}\right)$$

(a) find $\frac{\mathrm{d}y}{\mathrm{d}x}$, giving your answer in its simplest form. **[2]**

(b) Use integration by parts to find

$$\int \arcsin\left(\frac{x-1}{3}\right) \mathrm{d}x$$ **[4]**

14.

$$g(\theta) = (1 + \sin\theta - \cos\theta)^2$$

(a) Show that $g(\theta) = 2 + 2\sin\theta - 2\cos\theta - \sin 2\theta$. **[2]**

(b) Hence, using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \theta g(\theta)\,\mathrm{d}\theta$$ **[5]**

15. **(i)** Sketch the graph of $y = \arctan x$ for $x \in \mathbb{R}$, indicating any horizontal asymptotes. **[1]**

(ii) Prove that

$$\frac{\mathrm{d}}{\mathrm{d}x}(\arctan x) = \frac{1}{1+x^2} \quad \textbf{[3]}$$

(iii) Using integration by parts and suitable substitutions, show that

$$\int_0^1 x(\arctan x)^2 \,\mathrm{d}x = \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2}\ln 2 \quad \textbf{[8]}$$